P600 | Duplex seawater pump

Water-lubricated axial-piston technology, selected for saline-fluid duty.

Duplex construction supports the seawater application; it does not replace a materials review. Specify salinity, temperature, dissolved solids, cleaning chemicals and standby/flushing conditions along with flow and pressure.

Published parameter	390 cm³/rev	444 cm³/rev	625 cm³/rev
Maximum boosted speed (rpm)	1250	1250	1250
Maximum input power (kW)	72.3	82.3	115.8
Maximum flow (m³/h)	29.2	33.3	46.8
Maximum continuous pressure (bar)	80	80	80
Mass (kg)	120	120	120
Maximum temperature (°C)	50	50	50
Minimum temperature (°C)	2	2	2

Flow equivalents (US gpm): 128.56, 146.62, 206.05. 80 bar = 1,160 psi; 2-50 °C = 35.6-122 °F. Each column is a displacement option; maxima are independent ratings.

Design & environmental advantages

Product-water lubrication removes the pump oil circuit and its contamination pathway. A compact multi-piston arrangement produces smooth flow. Duplex materials address saline duty more deliberately than a generic freshwater specification.

Efficiency at the operating point

Compare input power with useful hydraulic output at the same displacement, speed and pressure. The supplied flow, power, torque and efficiency charts identify those conditions; they are not interchangeable with the 160 bar freshwater range.

Feed, drainage and installation

Published feed-pressure guide: 3.5-4.1 bar (51-59 psi), depending on shaft speed. Maintain inlet pressure at least 2.5 bar (36 psi) above case pressure. Separate internal leakage to a low-pressure drain. Specify coupling, rotation, supports, filtration, start-up flushing and service access.

Quotation and technical submittal

Provide flow, discharge pressure, inlet pressure at the pump, speed, feed analysis, duty cycle and site conditions. Request the selected displacement, wetted materials and seals, approved drawing, drive selection, pressure-test records and maintenance schedule. Match corrosion claims to the actual chemistry.

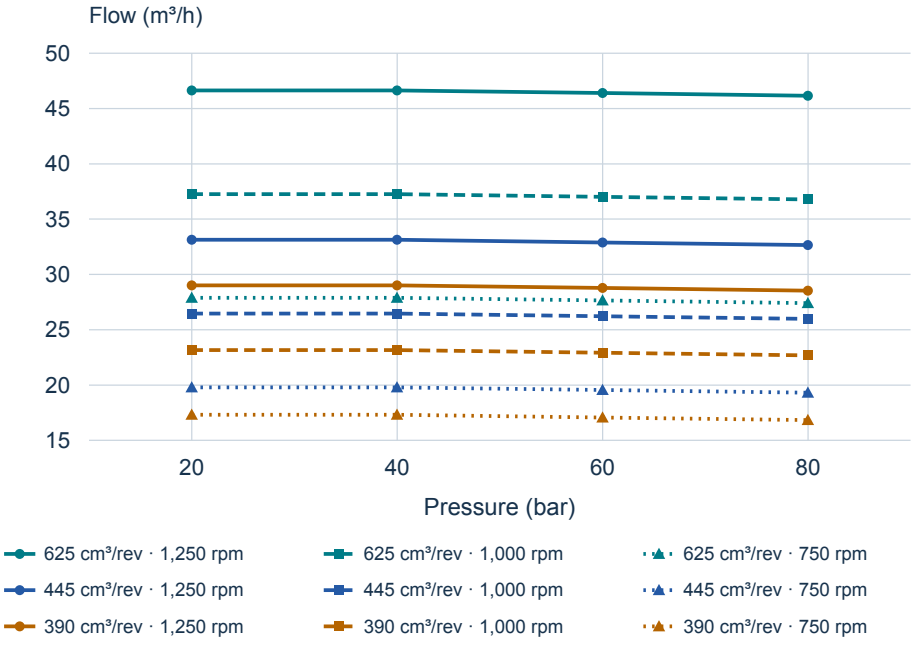
Source conflict: the table lists 444 cm³/rev, while some curve legends say 445. Both source entries remain identified; resolve the ordered displacement with the supplier.

Sea Water Axial Piston Pumps V.04.26 (3).pdf, pp. 2-4, 25-27.

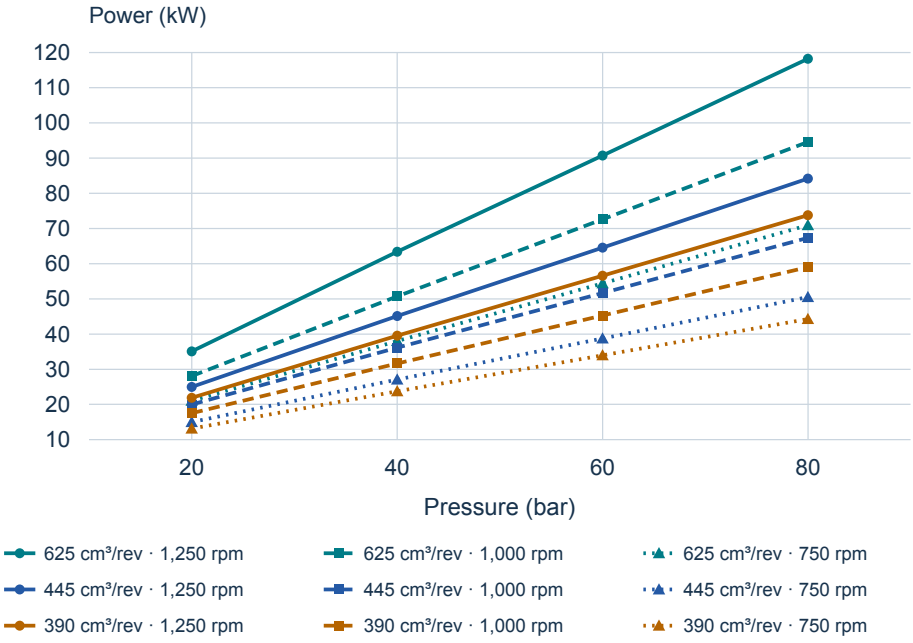
P600 | Performance

Read the source legend and stated operating conditions for each chart.

Output flow



Input power



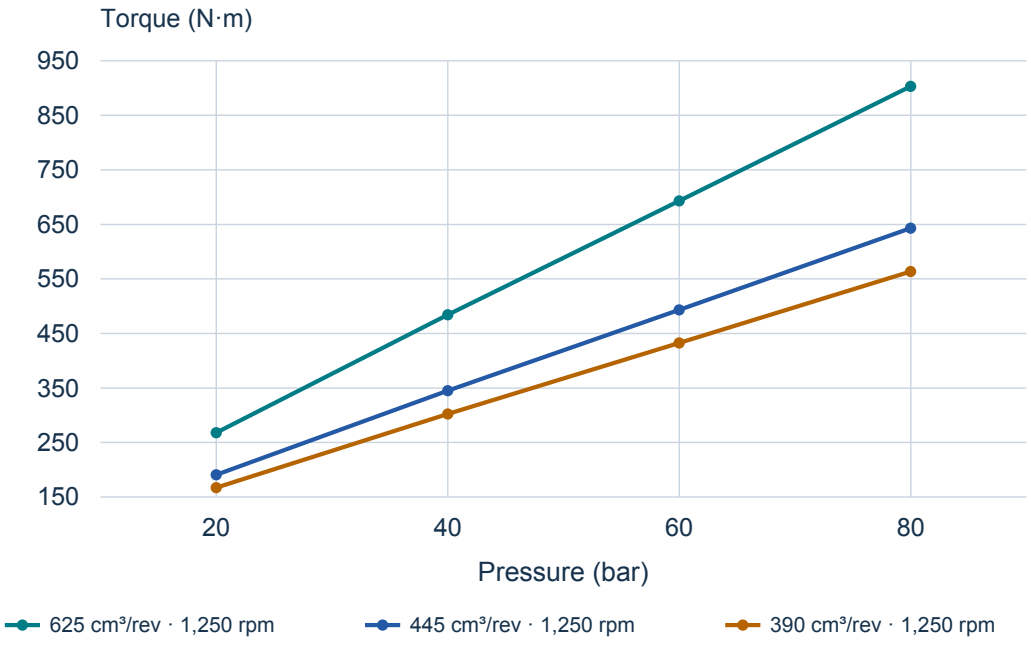
Colours identify displacement; line patterns and markers distinguish shaft speed. Native curve geometry was checked against the manufacturer PDF. 1 m³/h = 4.40287 US gpm; 1 N·m = 0.737562 lbf·ft.

Sea Water Axial Piston Pumps V.04.26 (3).pdf, pp. 26, 27.

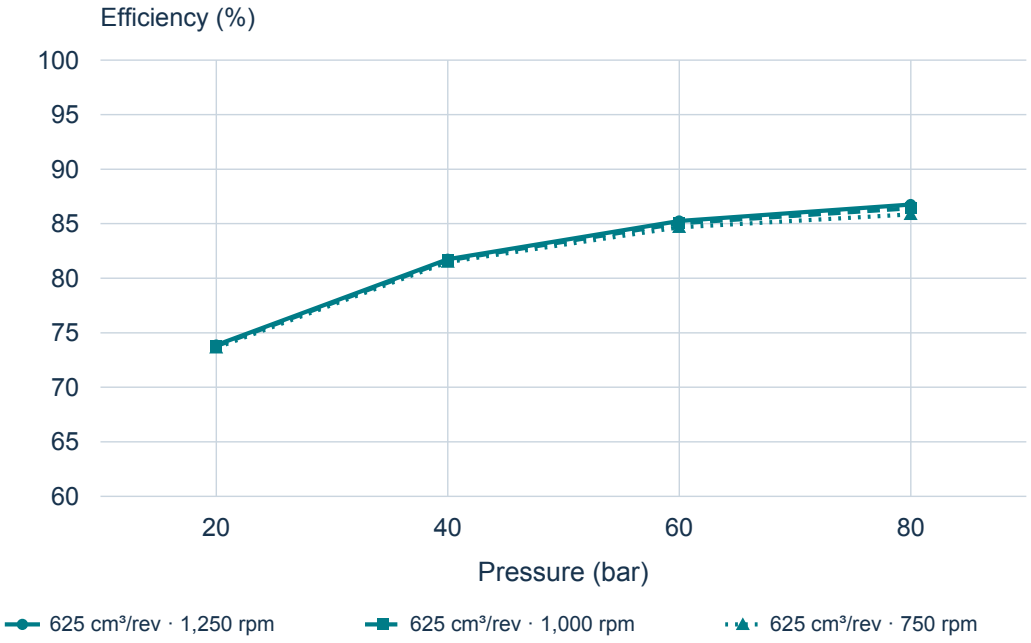
P600 | Performance

Read the source legend and stated operating conditions for each chart.

Input torque



Overall efficiency



Colours identify displacement; line patterns and markers distinguish shaft speed. Native curve geometry was checked against the manufacturer PDF. 1 m³/h = 4.40287 US gpm; 1 N·m = 0.737562 lbf·ft.

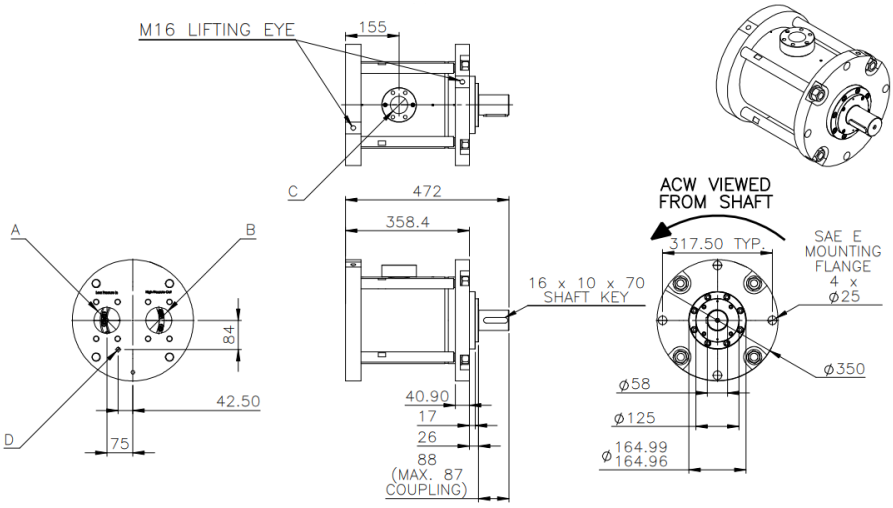
Sea Water Axial Piston Pumps V.04.26 (3).pdf, pp. 26, 27.

P600 | Dimensions & ports

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

SPECIFICATION				
Displacement (cc/rev)		390	444	625
Max. RPM	Boosted: *	1250	1250	1250
Max. Input Power (kW)		72.3	82.3	115.8
Max. Water Flow (m³/hr)		29.2	33.3	46.8
Max. Cont. Pressure (Barg)		80	80	80
Weight (kg)		120	120	120
Temperature (°C)	Max: **	50	50	50
	Min: ***	2	2	2

3.7.2 General Assembly Drawing



3.7.3 Port Dimensions (ACW Configuration)

Position	Description	Size
A	Low Pressure In	3" 3000 PSI CLASS SAE 4-BOLT FLANGE
B	High Pressure Out	3" 3000 PSI CLASS SAE 4-BOLT FLANGE
C	Case Drain	1 1/2" 3000 PSI CLASS SAE 4-BOLT FLANGE
D	Inlet Pressure Switch Port	1/4" BSPP FEMALE

The manufacturer specification and drawing are reproduced together to preserve port labels and configuration context. Nominal connection and flange ratings do not change the 80 bar pump operating limit.

Sea Water Axial Piston Pumps V.04.26 (3).pdf, p. 25.